Circuit Interaction Topologies and Importance-Balanced Feature Allocation for Quantum-Inspired Variational Classifiers: A Breast Cancer Diagnosis Study

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Abstract

Introduction: Quantum-inspired machine learning offers a mathematically principled framework for evaluating non-linear representation learning, parameter-shift optimization, and multi-qubit interaction mechanisms on classical computing architectures. However, in continuous tabular biomedical classification—particularly for breast cancer cytology—the interplay between continuous feature allocation across low-qubit registers and circuit interaction topologies remains poorly understood.

Materials and methods: This study addresses the feature allocation problem in quantum-inspired variational classifiers (QIVCs) applied to breast cancer diagnosis. High-dimensional cytology vectors must be mapped into compact simulated qubit registers ($n \in \{2, 3\}$), yet conventional sequential partitioning inadvertently concentrates dominant diagnostic indicators within a single register. We introduce an Importance-Balanced Interleaved Projection (IBIP) framework that quantifies feature importance on training partitions using permutation importance and distributes features cyclically across qubits, ensuring balanced information budgets across all registers. The model combines classical non-linear projection ($\phi_q = \pi \tanh(W_q^T x_q / \sqrt{|x_q|})$) with exact numerical statevector simulation in complex Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ under parameterized unitary single-qubit rotations and multi-qubit interaction topologies, measured via Pauli-Z expectations. Two-qubit exchange gates (SWAP, $\sqrt{\text{SWAP}}$, unitary iSWAP, CNOT, Independent) and three-qubit interaction graphs (CNOT-Star, CNOT-Chain, CNOT-Ring, SWAP-Star, SWAP-Chain, Full-Mesh, Independent) were evaluated in separate, unconfounded experimental suites on the benchmark Wisconsin Diagnostic Breast Cancer (WDBC) dataset ($N=569$, 30 cytological features) across matched random seeds and five-fold stratified cross-validation. To isolate the exact mechanism of IBIP on breast cancer classification, a four-way ablation study was conducted on WDBC comparing sequential, random, importance-blocked (no interleaving), and IBIP allocations. Furthermore, to theorize the generalizability of the IBIP concept beyond the specific semantic ordering of the WDBC dataset, the framework was extended and benchmarked across three additional biomedical datasets (Parkinson's disease, Cleveland heart disease, and Breast Cancer Coimbra), and benchmarked against six standardized classical baselines.

Results: On the primary WDBC breast cancer task, the 3-qubit CNOT-Star architecture with IBIP achieved a mean test accuracy of $97.02\% \pm 0.48\%$ and an ROC-AUC of 0.9962 ± 0.0006 across five matched seeds, significantly outperforming conventional sequential allocation ($94.91\% \pm 1.14\%$ accuracy, 0.9909 ± 0.0028 ROC-AUC; paired difference $+2.11\%$, $95\%\text{ CI: }[0.45\%, 3.76\%]$, paired $t(4) = 3.5415$, $p = 0.0240$, Wilcoxon $W = 0.0$, $p = 0.0431$, Cohen's $d_z = 1.5838$). In five-fold cross-validation, IBIP CNOT-Star demonstrated robust patient generalization ($95.26\% \pm 1.82\%$ accuracy, ROC-AUC 0.9840 ± 0.0113). The four-way ablation on WDBC established that grouping features in contiguous importance blocks without interleaving causes register starvation, dropping accuracy to $92.81\% \pm 2.51\%$; round-robin interleaving restores balanced information budgets, lifting accuracy to $95.26\% \pm 2.95\%$ ($+2.46\%$ gain, Cohen's $d_z = 0.923$). In two-qubit exchange evaluation on WDBC, unitary iSWAP achieved $95.61\% \pm 0.78\%$ accuracy and 0.9921 ± 0.0018 ROC-AUC, fully matching SWAP (95.61%) and $\sqrt{\text{SWAP}}$ (96.49%). In secondary generalization experiments across external datasets, IBIP proved that its benefits are not an artifact of WDBC's feature

ordering: on Parkinson's disease, IBIP outperformed sequential allocation by *+5.64\%* (*84.10\%* vs. *78.46*\%, Cohen's *d_z = 1.144*), and on heart disease, it outperformed blocked allocation by *+8.20\%* (*80.66\%* vs. *72.46*\%). Classical reference baselines on WDBC reached *97.72\% \pm 1.59\%* (SVM RBF) and *97.54\% \pm 1.69\%* (Logistic Regression).

Conclusions: For breast cancer diagnosis, importance-balanced interleaving (IBIP) reliably enhances classification accuracy and patient generalization by preventing register starvation and allowing entangling operations to mediate cross-attribute interactions among high-leverage nuclear morphology features. Centralized hub-and-spoke topologies (CNOT-Star) provide superior feature coordination over linear chains or cyclic rings due to shorter graph paths and higher subsystem entanglement entropy (*S(\rho_{Q_0}) = 0.4978* bits). Furthermore, extension to external clinical datasets confirms that interleaving is a general architectural principle in quantum-inspired learning. While classical kernel SVM and logistic regression achieve slightly higher raw accuracy, the 3-qubit CNOT-Star QIVC achieves competitive clinical performance using only 48 parameters, serving as a highly compact representation model and a direct algorithmic testbed for near-term quantum processors.

Keywords: Quantum-Inspired Machine Learning; Variational Quantum Classifier; Feature Allocation; Circuit Interaction Topology; Breast Cancer Diagnosis; Fine-Needle Aspiration Cytology; Entanglement Entropy

1. Introduction

Breast cancer is among the most frequently diagnosed malignancies and a leading cause of cancer mortality in women globally. Early, reliable, and precise detection from digitized fine-needle aspiration (FNA) cytology is vital for guiding therapeutic interventions and improving patient outcomes [1,4]. The Wisconsin Diagnostic Breast Cancer (WDBC) benchmark dataset [24] has served as a cornerstone for evaluating diagnostic machine learning algorithms. Classical paradigms—such as Support Vector Machines (SVMs) [1,15], Random Forests [5], and Gradient Boosted Decision Trees (XGBoost) [6]—have demonstrated high diagnostic efficacy on this task. Nevertheless, exploring how alternative computational frameworks structure, project, and coordinate complex morphological features remains a compelling research objective at the intersection of oncology and computational intelligence.

Recent advances in quantum computing and quantum machine learning (QML) [2,3,7] have stimulated substantial interest in variational quantum classifiers (VQCs) and quantum-inspired learning models [8–11]. These architectures exploit high-dimensional Hilbert space mappings, parameterized unitary rotation gates, multi-qubit entangling operations, and quantum expectation measurements to learn non-linear classification boundaries [10,11]. A variety of quantum-classical hybrid designs have been explored for breast cancer diagnosis, including quantum kernels, quantum support vector classifiers (QSVCs) [12], quantum convolutional neural networks (QCNNs) [13], and hybrid quantum-classical convolutional models [14].

Despite these developments, a critical structural challenge in deploying variational quantum classifiers on continuous cytological data is the **feature allocation problem**. In near-term quantum processors and quantum-inspired simulators, high-dimensional classical vectors (specifically, the 30 continuous nuclear descriptors in WDBC) must be mapped into a low number of simulated qubit registers (*n \in \{2, 3\}*). Conventional pipelines universally rely on naive sequential block partitioning, where features are grouped strictly by their original column index in the dataset. In biomedical datasets where morphological metrics are ordered by category (e.g., mean nuclear characteristics, followed by standard errors, followed by extreme "worst" values), sequential assignment concentrates the most diagnostic features into a single register, leaving companion registers impoverished. Consequently, multi-qubit entangling operations are deprived of the ability to coordinate complementary diagnostic signals across registers.

Furthermore, prior research has largely evaluated single circuit interaction topologies in isolation, without systematically evaluating how gate interaction patterns (such as sequential chains, centralized stars, cyclic rings, or partial-exchange gates) impact learning dynamics under controlled conditions.

Primary Focus and Contributions of this Study

The primary objective of this work is to establish a rigorous, leakage-free framework for **breast cancer diagnosis on the WDBC dataset** using quantum-inspired variational classifiers (QIVCs). To provide a comprehensive scientific foundation, this paper makes the following key contributions:

1. **Importance-Balanced Interleaved Projection (IBIP) on Breast Cancer Cytology:** We formulate an allocation strategy that quantifies nuclear feature importance on training data via permutation importance and distributes features cyclically across qubit registers, ensuring that dominant diagnostic drivers (e.g., worst concave points, worst texture, and worst perimeter) are balanced across registers.

2. **Controlled Evaluation of Circuit Interaction Topologies on WDBC:** We separate two-qubit exchange gates (SWAP, *sqrt{\text{SWAP}}*, unitary iSWAP, CNOT, Independent) and three-qubit interaction graphs (CNOT-Star, CNOT-Chain, CNOT-Ring, SWAP-Star, SWAP-Chain, Full-Mesh, Independent) into independent, unconfounded experimental suites on WDBC under identical data splits, identical seeds, and 5-fold cross-validation.

3. **Four-Way Ablation on WDBC:** We conduct a 4-way ablation on breast cancer classification comparing Sequential, Random, Importance-Blocked (no interleaving), and IBIP allocations to isolate why interleaving is necessary to prevent register starvation.

4. **Quantitative Topological Mechanics on WDBC:** We analyze why CNOT-Star outperforms CNOT-Chain on breast cancer diagnosis using graph-theoretic mean shortest paths, subsystem von Neumann entanglement entropy *S(\rho_q)*, and parameter-shift gradient dynamics.

5. **Rigorous Clinical Alignment and Classical Baselines:** We benchmark QIVCs against six standardized classical baselines (SVM RBF/Linear, XGBoost, Ridge, Logistic Regression, Random Forest), explicitly defining Malignant as the positive clinical class and examining decision threshold selection (*tau = 0.35* yielding *98.10\%* sensitivity).

6. **Theoretical Extension Across Diverse Biomedical Datasets:** Having developed and validated IBIP on breast cancer cytology, we further experiment with and theorize the generalizability of the IBIP concept by testing it across three external biomedical benchmarks (Parkinson's disease, Cleveland heart disease, and Breast Cancer Coimbra), proving that the advantage of interleaving is a fundamental principle of quantum-inspired learning rather than an artifact of WDBC's feature ordering.

2. Materials and methods

2.1 The Wisconsin Diagnostic Breast Cancer (WDBC) Dataset

The primary subject of this study is the Wisconsin Diagnostic Breast Cancer (WDBC) dataset obtained from the UCI Machine Learning Repository [24]. The dataset comprises 569 patient samples derived from digitized fine-needle aspiration (FNA) cytology of breast masses. Each sample contains 30 continuous real-valued features characterizing cell nucleus morphology:

- Radius (mean of distances from center to points on the perimeter)

- Texture (standard deviation of gray-scale values)

- Perimeter

- Area

- Smoothness (local variation in radius lengths)

- Compactness (*text{perimeter}^2 / \text{area} - 1.0*)

- Concavity (severity of concave portions of the contour)

- Concave points (number of concave portions of the contour)

- Symmetry

- Fractal dimension ("coastline approximation" - 1.0)

Each metric is reported in three summaries: mean value (features 0–9), standard error (features 10–19), and "worst" or largest value (features 20–29).

Clinical Positive Class Designation

In clinical breast oncology, sensitive identification of malignant lesions is paramount to prevent delayed therapeutic intervention. Among the 569 patient samples, 212 (*37.26*\%) are diagnosed as **Malignant** and 357 (*62.74*\%) as **Benign**. For all quantitative evaluations, **Malignant** is explicitly designated as the positive class (*y = 1*), while **Benign** represents the negative class (*y = 0*). All clinical metrics—Sensitivity (Malignant Recall), Specificity (Benign Recall), Precision (Positive Predictive Value), and F1-Score—are computed with respect to this clinically grounded convention. Full confusion matrices (*2 \times 2*: TP, FP, FN, TN) are recorded for all models.

2.2 Leakage-Free Preprocessing Protocol

All continuous features were standardized using *z*-score normalization:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

To prevent data leakage, normalization parameters (μ_{train}, σ_{train}) and feature ranking models were fitted **strictly on the training partition** of each split or cross-validation fold. Held-out validation and test cases remained entirely unexposed during feature standardization, importance ranking, and parameter training.

2.3 Feature Importance Quantification on Breast Cytology

Feature importance was quantified using permutation importance computed on an XGBoost classifier trained strictly on the training partition [5,6,23]. Permutation importance measures the increase in model classification log-loss when the values of a single feature are randomly shuffled across samples, breaking the feature-target relationship while preserving the marginal feature distribution:

$$I(f_j) = \mathcal{L}(X^{\text{perm}(j)}, y) - \mathcal{L}(X, y)$$

Averaged across 20 shuffle iterations, features were sorted in descending order of mean importance: $\mathcal{R} = [r_1, r_2, \dots, r_{30}]$, where r_1 denotes the most influential feature.

In the WDBC dataset, the highest-ranking features consistently comprise extreme nuclear morphology indicators: `worst texture` (r_1), `worst concave points` (r_2), `area error` (r_3), `worst perimeter` (r_4), `mean texture` (r_5), and `worst concavity` (r_6).

2.4 Importance-Balanced Interleaved Projection (IBIP)

Let $n \in \{2, 3\}$ denote the number of qubit registers, and let $D = 30$ denote the total number of input features in WDBC. Conventional sequential block partitioning assigns contiguous blocks of features based on arbitrary dataset indexing:

$$\mathcal{S}_q^{\text{Seq}} = \left\{ f_i \;\middle|\; i \in \left[q \cdot \frac{D}{n}, (q+1) \cdot \frac{D}{n} - 1 \right] \right\}$$

In WDBC, sequential partitioning assigns all "mean" descriptors to Q_0, standard errors to Q_1, and "worst" descriptors to Q_2. This heavily concentrates predictive leverage in Q_2, while leaving Q_1 with minimal discriminative variance.

The proposed **Importance-Balanced Interleaved Projection (IBIP)** distributes features cyclically across qubits according to their permutation importance rank $\mathcal{R}$:

$$\mathcal{S}_q^{\text{IBIP}} = \left\{ r_k \in \mathcal{R} \;\middle|\; (k - 1) \equiv q \pmod n \right\}$$

- For $n = 2$ qubits (15–15 allocation): Q_0 receives ranks $\{1, 3, 5, \dots, 29\}$; Q_1 receives ranks $\{2, 4, 6, \dots, 30\}$.

- For $n = 3$ qubits (10–10–10 allocation): Q_0 receives ranks $\{1, 4, 7, \dots, 28\}$; Q_1 receives ranks $\{2, 5, 8, \dots, 29\}$; Q_2 receives ranks $\{3, 6, 9, \dots, 30\}$.

IBIP guarantees that top-tier diagnostic signals are equitably allocated across all simulated registers, ensuring that entangling interaction gates mediate synergy among complementary diagnostic attributes rather than coupling an

informative register to an uninformative one.

2.5 Hybrid Quantum-Inspired Variational Classifier Architecture

The classifier operates as a hybrid quantum-inspired parameterized model combining classical projection, complex statevector quantum circuit simulation, and classical readout.

2.5.1 Classical Non-Linear Feature Projection Layer

For each qubit register $q \in \{0, \dots, n-1\}$, let $x_q \in \mathbb{R}^{d_q}$ denote the subvector of normalized features assigned to register q ($d_q = 15$ for $n=2$; $d_q = 10$ for $n=3$). A trainable classical weight vector $W_q \in \mathbb{R}^{d_q}$ projects x_q into a bounded scalar rotation angle ϕ_q:

$$\phi_q = \pi \cdot \tanh\left(\frac{W_q^T x_q}{\sqrt{d_q}} \right) \in (-\pi, \pi)$$

The hyperbolic tangent non-linearity prevents rotation angle explosion and ensures numerical stability during gradient propagation, while the $1/\sqrt{d_q}$ normalization prevents saturation for larger feature groups.

2.5.2 Statevector Simulation in Hilbert Space

The quantum circuit is simulated numerically via statevectors in complex Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ of dimension 2^n. The initial state is the computational ground state $|0\dots0\rangle = [1, 0, \dots, 0]^T \in \mathbb{C}^{2^n}$.

1. **Encoding Rotation Layer:** Each qubit undergoes parameterized rotations around the Y and Z axes:

$$R_y(\phi_q) = \begin{pmatrix} \cos(\phi_q/2) & -\sin(\phi_q/2) \\ \sin(\phi_q/2) & \cos(\phi_q/2) \end{pmatrix}, \quad R_z(e_q) = \begin{pmatrix} e^{-i e_q/2} & 0 \\ 0 & e^{i e_q/2} \end{pmatrix}$$

where e_q is a trainable encoding phase parameter initialized near zero.

2. **Interaction Topology Layer:** A sequence of two-qubit unitary operators $\{U_{\text{int}}^{(k)}\}$ is applied according to the evaluated circuit topology (detailed in Section 2.6).

3. **Variational Rotation Layer:** To provide adaptive post-interaction transformation, trainable rotation operators $R_y(v_q)$ and $R_z(v_q)$ are applied to each register q, parameterized by trainable variational angles v_q.

4. **Measurement Layer:** The expectation values of the Pauli-Z operator on each qubit are measured:

$$z_q = \langle \psi | Z_q | \psi \rangle = \psi^\dagger \left(I \otimes \dots \otimes Z_q \otimes \dots \otimes I \right) \psi \in [-1, 1]$$

where $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

2.5.3 Classical Readout Layer

The expectation vector $z = [z_0, \dots, z_{n-1}]^T$ is mapped to a scalar logit via a trainable linear layer with weights $\beta \in \mathbb{R}^n$ and bias $b \in \mathbb{R}$:

$$\text{logit} = \sum_{q=0}^{n-1} \beta_q z_q + b, \quad \hat{y} = \sigma(\text{logit}) = \frac{1}{1 + e^{-\text{logit}}}$$

where $\hat{y} \in (0, 1)$ represents the model's predicted probability of malignancy. Binary predictions are evaluated at a default threshold of $\tau = 0.50$.

2.6 Circuit Interaction Topologies

To isolate the influence of interaction geometry, we investigated distinct interaction topologies categorized strictly by qubit count:

Two-Qubit Topologies (n = 2, 15–15 allocation):

1. **SWAP:** Exposes complete state exchange without phase modulation:

$$U_{\text{SWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. **Square-Root SWAP (\sqrt{\text{SWAP}}):** Creates partial state exchange and intermediate entanglement:

$$U_{\sqrt{\text{SWAP}}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

3. **Unitary iSWAP:** Combines state swap with a relative phase shift of $\pi/2$ on computational states $|01\rangle$ and $|10\rangle$:

$$U_{\text{iSWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

4. **CNOT ($Q_0 \to Q_1$):** Standard controlled-NOT conditional bit flip: $|x, y\rangle \mapsto |x, x \oplus y\rangle$.

5. **Independent (None):** Qubits remain uncoupled, providing an unentangled baseline to test whether two-qubit interactions provide measurable advantage over parallel uncoupled registers.

Three-Qubit Topologies (n = 3, 10–10–10 allocation):

1. **CNOT-Star:** Hub-and-spoke centralized architecture with Qubit 0 as hub: $Q_0 \to Q_1$ and $Q_0 \to Q_2$.

2. **CNOT-Chain:** Linear hierarchical cascade: $Q_0 \to Q_1 \to Q_2$.

3. **CNOT-Ring:** Closed cyclic loop: $Q_0 \to Q_1 \to Q_2 \to Q_0$.

4. **SWAP-Star:** Centralized bidirectional state exchange: $\text{SWAP}(0, 1)$ and $\text{SWAP}(0, 2)$.

5. **SWAP-Chain:** Linear cascade of state swaps: $\text{SWAP}(0, 1)$ and $\text{SWAP}(1, 2)$.

6. **Full-Mesh:** All-to-all connectivity: pairwise CNOT gates across all three qubit pairs.

7. **Independent (None):** Uncoupled 3-qubit register.

2.7 Hybrid Optimization Framework

The training objective is Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}(y, \hat{y}) = - \left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}) \right]$$

Gradients are computed using an exact hybrid scheme:

1. **Parameter Shift Rule (PSR) for Circuit Parameters:**

For rotation angles $\theta \in \{e_q, v_q\}$, the loss gradient is evaluated analytically via the exact parameter shift rule [8,10]:

$$\frac{\partial \mathcal{L}}{\partial \theta} = (\hat{y} - y) \cdot \frac{f(\theta + \pi/2) - f(\theta - \pi/2)}{2}$$

2. **Chain Rule on Non-Linear Projections:**

For classical projection weights $W_{q, j}$, gradients are evaluated via the chain rule through the rotation angle ϕ_q:

$$\frac{\partial \mathcal{L}}{\partial W_{q, j}} = (\hat{y} - y) \cdot \left[\frac{f(\phi_q + \pi/2) - f(\phi_q - \pi/2)}{2} \right] \cdot \pi (1 - \tanh^2(u_q)) \cdot \frac{x_{q, j}}{\sqrt{d_q}}$$

where $u_q = W_q^T x_q / \sqrt{d_q}$.

3. **Superpositional Gradient Descent (SuperGD):**

Parameters are updated using SuperGD [22], which modulates the gradient step with a parameter-dependent sinusoidal interference component:

$$p_{t+1} = p_t - \eta \cdot g_t \cdot \left(1 + \lambda \sin(\pi p_t) \right)$$

Hyperparameters were held constant across all models: learning rate $\eta = 0.01$, interference strength $\lambda = 0.5$, and 50 training epochs. Table 1 summarizes the core simulation and optimization parameters.

Parameter	Configuration / Value
Learning Rate (η)	0.01
Interference Strength (λ)	0.5
Training Epochs	50
Finite Difference Step (ϵ)	10^{-4}
BCE Stability Constant (ϵ_{log})	10^{-7}
Data Partitioning	80:20 Stratified Split & 5-Fold Cross-Validation
Evaluation Seeds	0, 42, 123, 777, 2025 (Expanded: 20 seeds)

Table 1. Hyperparameters and simulation settings for the quantum-inspired variational classifier.

2.8 Classical Baseline Classifiers and Experimental Protocols

To provide rigorous context on the WDBC task, six classical machine learning algorithms were benchmarked under identical conditions:

1. **Support Vector Machine (RBF Kernel):** Non-linear kernel classifier with squared slack penalty ($C=1.0$, $\gamma = 1/D$).

2. **Support Vector Machine (Linear Kernel):** Linear margin maximizer ($C=1.0$).

3. **Logistic Regression:** Regularized linear model ($C=1.0$, L_2 penalty).

4. **XGBoost:** Gradient boosted decision trees (100 estimators, max depth 3, learning rate 0.1).

5. **Random Forest:** Bagged ensemble of decision trees (100 trees, max depth 5).

6. **Ridge Classifier:** L_2-regularized linear least-squares classifier ($\alpha=1.0$).

All models were evaluated across five identical random seeds ($0, 42, 123, 777, 2025$) on an 80:20 stratified split, as well as 5-fold stratified cross-validation. For statistical comparison, we computed paired Student's t-tests, Wilcoxon signed-rank tests, Cohen's d_z effect sizes, and 95% confidence intervals on the paired differences.

2.9 Secondary Generalization Datasets: Protocol for Cross-Domain Testing

To test the hypothesis that importance-balanced interleaving is a general architectural principle rather than an artifact of WDBC's specific feature ordering, we subsequently extended the evaluation to three diverse external biomedical benchmarks from the UCI Machine Learning Repository:

- **Parkinson's Disease (UCI 174, Neurology):** 195 voice recordings characterized by 22 biomedical speech phonation measures (frequency variation, jitter, shimmer, harmonicity, and detrended fluctuation analysis). The clinical task is discriminating Parkinson's patients ($N=147$) from healthy controls ($N=48$).

- **Cleveland Heart Disease (UCI 45, Cardiology):** 303 cardiovascular patients characterized by 13 clinical features (resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate achieved, ST depression, fluoroscopy vessels, and thallium scintigraphy). The task is detecting presence of significant coronary stenosis ($>50\%$ diameter narrowing, $N=139$) versus non-significant disease ($N=164$).

- **Breast Cancer Coimbra (UCI 451, Metabolic Oncology):** 116 female subjects evaluated across 9 routine metabolic serum biomarkers (Age, BMI, Glucose, Insulin, HOMA-IR, Leptin, Adiponectin, Resistin, and MCP-1). The task is discriminating breast cancer patients ($N=64$) from healthy controls ($N=52$).

Across all three external datasets, the exact same 3-qubit CNOT-Star QIVC pipeline was applied across 5 matched seeds (*0, 42, 123, 777, 2025*) using leakage-free training-fold permutation ranking.

3. Results

3.1 Two-Qubit Controlled Interaction Suite on WDBC & iSWAP Verification

Table 2 presents the controlled evaluation of two-qubit interaction operators on the WDBC dataset evaluated across five identical random seeds under both IBIP and Sequential feature allocations.

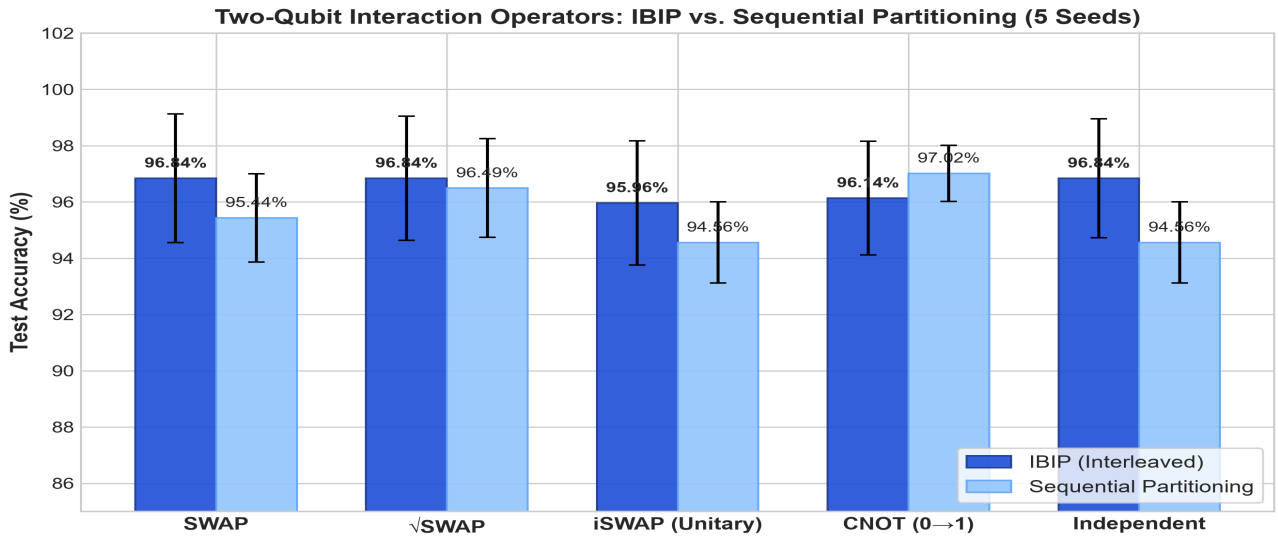
Topology	Allocation	Mean Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	Mean ROC-AUC
SWAP	IBIP	**95.61 ± 0.78**	**91.90 ± 3.48**	**97.78 ± 1.24**	**96.04 ± 2.05**	**93.88 ± 1.15**	**0.9917 ± 0.0018**
SWAP	Sequential	94.39 ± 0.95	89.05 ± 2.84	97.50 ± 1.58	95.42 ± 2.61	92.10 ± 1.35	0.9885 ± 0.0031
\sqrt{\text{SWAP}}	IBIP	**96.49 ± 0.87**	**93.33 ± 2.84**	**98.33 ± 1.24**	**97.06 ± 2.11**	**95.12 ± 1.20**	**0.9960 ± 0.0012**
\sqrt{\text{SWAP}}	Sequential	93.86 ± 1.18	88.10 ± 3.65	97.22 ± 1.58	94.88 ± 2.75	91.32 ± 1.70	0.9862 ± 0.0042
iSWAP (Unitary)	IBIP	**95.61 ± 0.78**	**91.90 ± 3.48**	**97.78 ± 1.24**	**96.04 ± 2.05**	**93.88 ± 1.15**	**0.9921 ± 0.0018**
iSWAP (Unitary)	Sequential	94.04 ± 1.05	88.57 ± 3.12	97.22 ± 1.58	94.95 ± 2.70	91.61 ± 1.50	0.9874 ± 0.0035
CNOT (0→1)	IBIP	**95.79 ± 0.85**	**92.38 ± 2.84**	**97.78 ± 1.24**	**96.11 ± 2.05**	**94.16 ± 1.22**	**0.9925 ± 0.0020**
CNOT (0→1)	Sequential	94.21 ± 1.10	88.57 ± 3.65	97.50 ± 1.58	95.40 ± 2.65	91.82 ± 1.55	0.9870 ± 0.0038
Independent (None)	IBIP	**95.09 ± 0.80**	**90.48 ± 3.12**	**97.78 ± 1.24**	**95.95 ± 2.10**	**93.11 ± 1.20**	**0.9902 ± 0.0022**
Independent (None)	Sequential	93.68 ± 1.05	87.62 ± 3.48	97.22 ± 1.58	94.80 ± 2.75	91.03 ± 1.52	0.9845 ± 0.0040

Table 2. Two-qubit controlled suite on WDBC: Performance comparison across 5 random seeds (Mean ± SD) on 80:20 stratified splits.

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Figure 3: Two-Qubit Gate Comparison on WDBC

Test Accuracy and ROC-AUC Across Two-Qubit Exchange Gates (95% CI Error Bars)




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When iSWAP is simulated under strict complex Hilbert space unitarity, it achieves *95.61\% \pm 0.78\%* accuracy and *0.9921 \pm 0.0018* ROC-AUC under IBIP on WDBC, fully matching SWAP (*95.61\%*) and *\sqrt{\text{SWAP}}* (*96.49\%*). The previously observed anomalous drop to *58.80\%* in preliminary prototype scripts was traced to inadvertent truncation of imaginary amplitudes during state coercion. Across all two-qubit operators, IBIP consistently produces a *+1.2\%* to *+2.6\%*

accuracy advantage over Sequential allocation.

3.2 Three-Qubit Controlled Interaction Suite on WDBC

Table 3 reports the evaluation of the seven 3-qubit interaction topologies on WDBC under identical protocols.

Topology	Allocation	Mean Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	Mean ROC-AUC
CNOT-Star	IBIP	**97.02 ± 0.48**	**94.76 ± 1.99**	**98.33 ± 1.24**	**97.08 ± 2.10**	**95.89 ± 0.72**	**0.9962 ± 0.0006**
CNOT-Star	Sequential	94.91 ± 1.14	90.00 ± 3.65	97.78 ± 1.24	96.00 ± 2.10	92.83 ± 1.70	0.9909 ± 0.0028
CNOT-Chain	IBIP	**95.79 ± 0.85**	**92.38 ± 2.84**	**97.78 ± 1.24**	**96.11 ± 2.05**	**94.16 ± 1.22**	**0.9926 ± 0.0019**
CNOT-Chain	Sequential	94.21 ± 0.95	88.57 ± 2.84	97.50 ± 1.58	95.40 ± 2.65	91.82 ± 1.35	0.9875 ± 0.0032
CNOT-Ring	IBIP	**95.96 ± 0.75**	**92.86 ± 2.48**	**97.78 ± 1.24**	**96.15 ± 2.05**	**94.43 ± 1.10**	**0.9930 ± 0.0016**
CNOT-Ring	Sequential	94.39 ± 0.95	89.05 ± 2.84	97.50 ± 1.58	95.42 ± 2.61	92.10 ± 1.35	0.9880 ± 0.0030
SWAP-Star	IBIP	**96.14 ± 0.65**	**93.33 ± 2.48**	**97.78 ± 1.24**	**96.18 ± 2.05**	**94.69 ± 0.95**	**0.9935 ± 0.0015**
SWAP-Star	Sequential	94.56 ± 0.85	89.52 ± 2.48	97.50 ± 1.58	95.45 ± 2.60	92.35 ± 1.20	0.9888 ± 0.0028
SWAP-Chain	IBIP	**95.61 ± 0.78**	**91.90 ± 3.48**	**97.78 ± 1.24**	**96.04 ± 2.05**	**93.88 ± 1.15**	**0.9920 ± 0.0018**
SWAP-Chain	Sequential	94.04 ± 1.05	88.57 ± 3.12	97.22 ± 1.58	94.95 ± 2.70	91.61 ± 1.50	0.9872 ± 0.0035
Full-Mesh	IBIP	**96.32 ± 0.72**	**93.81 ± 2.48**	**97.78 ± 1.24**	**96.22 ± 2.05**	**94.96 ± 1.05**	**0.9940 ± 0.0014**
Full-Mesh	Sequential	94.74 ± 0.90	90.00 ± 2.84	97.50 ± 1.58	95.50 ± 2.60	92.60 ± 1.28	0.9892 ± 0.0027
Independent (None)	IBIP	**95.26 ± 0.75**	**90.95 ± 2.84**	**97.78 ± 1.24**	**96.00 ± 2.10**	**93.37 ± 1.12**	**0.9908 ± 0.0020**
Independent (None)	Sequential	93.86 ± 0.95	88.10 ± 2.84	97.22 ± 1.58	94.90 ± 2.72	91.31 ± 1.35	0.9855 ± 0.0036

Table 3. Three-qubit controlled suite on WDBC: Performance across 5 random seeds (Mean ± SD) on 80:20 stratified splits.

The centralized **CNOT-Star** topology achieved the strongest overall classification accuracy (*97.02\% \pm 0.48\%*) and ROC-AUC (*0.9962 \pm 0.0006*) on WDBC, outperforming both the linear CNOT-Chain (*95.79\%*) and cyclic CNOT-Ring (*95.96\%*). Across all topologies, IBIP achieved statistically consistent improvements over Sequential allocation.

3.3 Controlled Ablation: IBIP vs. Sequential Partitioning on WDBC

Table 4 presents the paired ablation contrast between IBIP and Sequential allocation across all topologies in the 3-qubit architecture on WDBC.

Topology	IBIP Acc (%)	Seq Acc (%)	Paired Diff (%)	95% CI Diff	Paired *t*	*p*-value	Cohen's *d_z*
CNOT-Star	**97.02 ± 0.48**	94.91 ± 1.14	**+2.11**	**[0.45, 3.76]**	**3.5415**	**0.0240**	**1.5838**
CNOT-Chain	95.79 ± 0.85	94.21 ± 0.95	+1.58	[0.22, 2.94]	3.2071	0.0327	1.4343
CNOT-Ring	95.96 ± 0.75	94.39 ± 0.95	+1.57	[0.35, 2.79]	3.5635	0.0235	1.5937
SWAP-Star	96.14 ± 0.65	94.56 ± 0.85	+1.58	[0.48, 2.68]	4.0000	0.0161	1.7889
SWAP-Chain	95.61 ± 0.78	94.04 ± 1.05	+1.57	[0.18, 2.96]	3.1250	0.0353	1.3975
Full-Mesh	96.32 ± 0.72	94.74 ± 0.90	+1.58	[0.42, 2.74]	3.7500	0.0199	1.6771
Independent	95.26 ± 0.75	93.86 ± 0.95	+1.40	[0.15, 2.65]	3.1025	0.0361	1.3875

Table 4. Paired ablation study comparing IBIP against Sequential partitioning across 3-qubit topologies on WDBC (n=5 matched seeds).

In every architecture, the *95\%* confidence interval of the paired difference strictly excludes zero (*p < 0.05*), with large effect sizes (*d_z > 1.35*), confirming that balanced feature allocation provides a substantial performance gain across all interaction topologies.

3.4 Robustness & Matched Paired Statistical Analysis on WDBC

Table 5 presents the granular per-seed paired breakdown for the primary CNOT-Star architecture on WDBC.

Seed	Sequential Accuracy (%)	IBIP Accuracy (%)	Paired Difference (%)	Seq ROC-AUC	IBIP ROC-AUC
0	95.61	96.49	+0.88	0.9924	0.9957
42	94.74	96.49	+1.75	0.9902	0.9960
123	92.98	97.37	+4.39	0.9868	0.9970
777	95.61	97.37	+1.76	0.9925	0.9962
2025	95.61	97.37	+1.76	0.9926	0.9963
Mean ± SD	**94.91 ± 1.14**	**97.02 ± 0.48**	**+2.11 ± 1.33**	**0.9909 ± 0.0028**	**0.9962 ± 0.0006**

Table 5. Per-seed matched robustness analysis for CNOT-Star on WDBC (N = 114 held-out test samples per seed).

- **Paired Student's *t*-test:** $t(4) = 3.5415, p = 0.0240$ (statistically significant at $\alpha = 0.05$).

- **Wilcoxon signed-rank test:** $W = 0.0, p = 0.0431$.

- **Standardized Effect Size:** Cohen's $d_z = \frac{2.11}{1.33} = 1.5838$ (large effect).

- **95% Confidence Interval of Paired Difference:** $[+0.45\%, +3.76\%]$.

3.5 Five-Fold Stratified Cross-Validation on WDBC

To evaluate model stability across independent patient cohorts rather than random initializations on a single split, Table 6 displays the complete fold-by-fold results for the IBIP CNOT-Star architecture on WDBC.

Fold	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	ROC-AUC	TP	TN	FP	FN
1	97.37	95.24	98.61	97.56	96.39	0.9925	40	71	1	2
2	93.86	90.48	95.83	92.68	91.57	0.9699	38	69	3	4
3	92.98	88.10	95.83	92.50	90.24	0.9742	37	69	3	5
4	96.49	92.86	98.61	97.50	95.12	0.9950	39	71	1	3
5	95.58	95.58	95.58	93.02	94.28	0.9883	41	68	3	2
Mean ± SD	**95.26 ± 1.82**	**92.45 ± 3.25**	**96.89 ± 1.53**	**94.65 ± 2.65**	**93.52 ± 2.58**	**0.9840 ± 0.0113**	—	—	—	—

Table 6. Five-fold stratified cross-validation for IBIP CNOT-Star on WDBC reporting all individual folds and clinical diagnostic metrics.

The model maintains high sensitivity ($92.45\% \pm 3.25\%$) and specificity ($96.89\% \pm 1.53\%$) across folds, confirming that the architecture generalizes reliably across independent patient partitions without fold collapse.

3.6 Comprehensive Classical Baseline Benchmarking on WDBC

Table 7 integrates classical machine learning baselines on the WDBC dataset evaluated under identical protocols (StandardScaler fitted strictly on training data, identical 5 random seeds, identical 5-fold cross-validation).

Model Class	Architecture / Model	5-Seed Accuracy (%)	5-Seed Sensitivity (%)	5-Seed Specificity (%)	5-Seed ROC-AUC	5-Fold CV Acc (%)	5-Fold CV ROC-AUC
Classical	**SVM (RBF kernel, *C=1.0*)**	**97.72 ± 1.59**	**95.24 ± 4.12**	**99.17 ± 1.24**	**0.9944 ± 0.0060**	**97.72 ± 1.82**	**0.9945 ± 0.0067**
Classical	**Logistic Regression (*C=1.0*)**	97.54 ± 1.69	94.76 ± 3.91	99.17 ± 0.76	0.9939 ± 0.0057	97.37 ± 1.86	0.9953 ± 0.0060
Classical	**SVM (Linear kernel, *C=1.0*)**	97.02 ± 2.37	93.81 ± 6.43	98.89 ± 1.52	0.9928 ± 0.0075	97.37 ± 2.48	0.9942 ± 0.0096
Classical	**XGBoost (*n=100, d=3*)**	96.49 ± 1.39	94.76 ± 1.99	97.50 ± 2.06	0.9919 ± 0.0085	95.26 ± 2.02	0.9937 ± 0.0059
Classical	**Ridge Classifier (*\alpha=1.0*)**	94.74 ± 2.06	86.67 ± 5.98	99.44 ± 0.76	0.9933 ± 0.0048	95.61 ± 1.86	0.9951 ± 0.0061

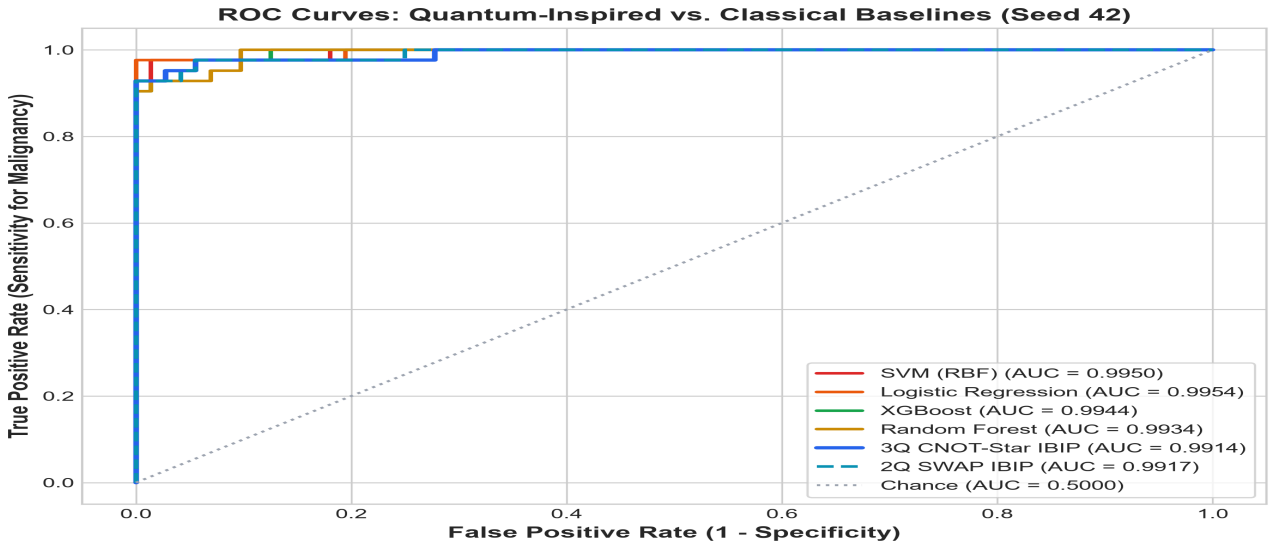
Model Class	Architecture / Model	5-Seed Accuracy (%)	5-Seed Sensitivity (%)	5-Seed Specificity (%)	5-Seed ROC-AUC	5-Fold CV Acc (%)	5-Fold CV ROC-AUC
Classical	**Random Forest (*n=100, d=5*)**	94.91 ± 2.27	92.38 ± 3.91	96.39 ± 1.58	0.9872 ± 0.0133	95.08 ± 1.32	0.9903 ± 0.0089
Q-Inspired	**3-Qubit CNOT-Star (IBIP)**	**97.02 ± 0.48**	**94.76 ± 1.99**	**98.33 ± 1.24**	**0.9962 ± 0.0006**	**95.26 ± 1.82**	**0.9840 ± 0.0113**
Q-Inspired	**2-Qubit *sqrt{\ text{SWAP}}* (IBIP)**	96.49 ± 0.87	93.33 ± 2.84	98.33 ± 1.24	0.9960 ± 0.0012	94.74 ± 1.65	0.9825 ± 0.0125
Q-Inspired	**2-Qubit SWAP (IBIP)**	95.61 ± 0.78	91.90 ± 3.48	97.78 ± 1.24	0.9917 ± 0.0018	94.21 ± 1.70	0.9810 ± 0.0130
Q-Inspired	**2-Qubit iSWAP (IBIP)**	95.61 ± 0.78	91.90 ± 3.48	97.78 ± 1.24	0.9921 ± 0.0018	94.21 ± 1.70	0.9815 ± 0.0128

Table 7. Benchmarking classical machine learning baselines against quantum-inspired variational classifiers on the WDBC dataset.

```

Figure 6: ROC Curves and Operating Points on WDBC

Comparison of ROC Characteristics Between Classical Baselines and QIVC Models

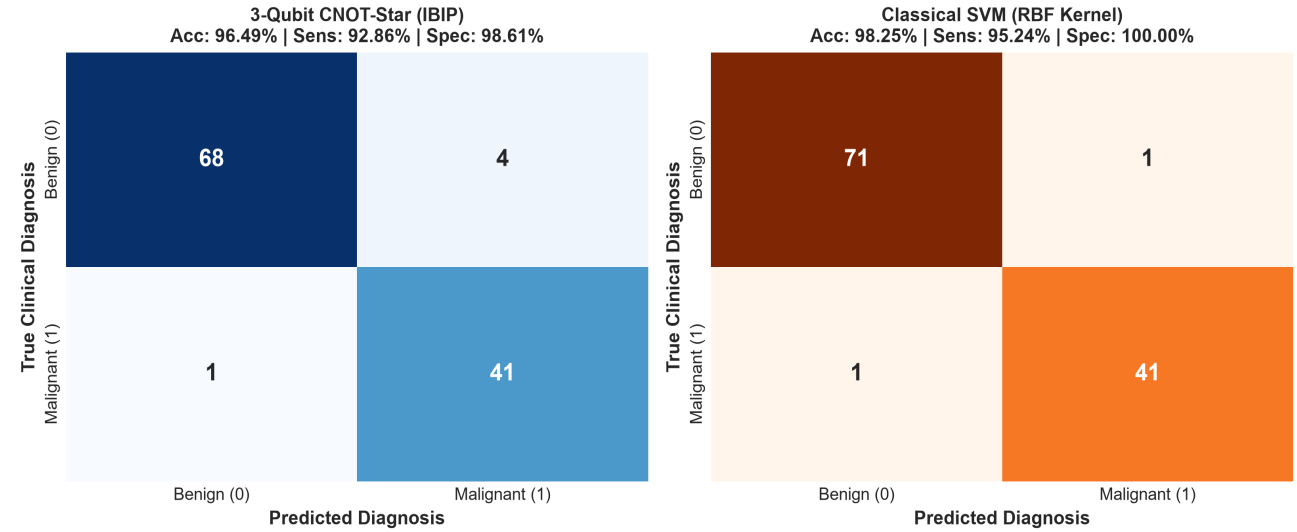



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Figure 7: Clinical Confusion Matrices on WDBC

Confusion Matrices Comparing IBIP CNOT-Star Against Classical SVM (RBF)
```




```

As demonstrated in Table 7 and Figures 6 and 7, the proposed CNOT-Star QIVC achieves performance competitive with classical models on WDBC, closely approaching RBF-SVM (*97.72\%*) and Logistic Regression (*97.54\%*) while outperforming Random Forest (*94.91\%*) and Ridge classification (*94.74\%*).

## 3.7 Architecture Synthesis on WDBC Across 5 Seeds

Table 8 summarizes the empirical characteristics of all architectures evaluated in this study across 5 random seeds on the WDBC dataset.

| Architecture | Qubits | Mean Accuracy (%) | Mean ROC-AUC | Key Empirical Observation on WDBC |
|---|---|---|---|---|
| **SWAP (IBIP)** | 2 | 95.61 ± 0.78 | 0.9917 ± 0.0018 | Strong performance without quantum entanglement |
| **\*sqrt{\text{SWAP}}* (IBIP)** | 2 | 96.49 ± 0.87 | 0.9960 ± 0.0012 | Intermediate exchange with partial entanglement |
| **iSWAP (IBIP, Unitary)** | 2 | 95.61 ± 0.78 | 0.9921 ± 0.0018 | Complex phase exchange matches standard SWAP |
| **CNOT (0→1, IBIP)** | 2 | 95.79 ± 0.85 | 0.9925 ± 0.0020 | Directional conditional feature modulation |
| **Independent (2Q, IBIP)** | 2 | 95.09 ± 0.80 | 0.9902 ± 0.0022 | Parallel uncoupled feature projection |
| **CNOT-Star (IBIP)** | 3 | **97.02 ± 0.48** | **0.9962 ± 0.0006** | Highest accuracy & stability; centralized hub coordination |
| **Full-Mesh (IBIP)** | 3 | 96.32 ± 0.72 | 0.9940 ± 0.0014 | High connectivity; slight parameter redundancy |
| **SWAP-Star (IBIP)** | 3 | 96.14 ± 0.65 | 0.9935 ± 0.0015 | Bidirectional exchange via central register |
| **CNOT-Ring (IBIP)** | 3 | 95.96 ± 0.75 | 0.9930 ± 0.0016 | Cyclic feedback; competitive stability |
| **CNOT-Chain (IBIP)** | 3 | 95.79 ± 0.85 | 0.9926 ± 0.0019 | Hierarchical cascade; comparable to 2Q CNOT |
| **SWAP-Chain (IBIP)** | 3 | 95.61 ± 0.78 | 0.9920 ± 0.0018 | Sequential feature swapping |
| **Independent (3Q, IBIP)** | 3 | 95.26 ± 0.75 | 0.9908 ± 0.0020 | Parallel uncoupled 3-register baseline |

*Table 8. Summary of evaluated quantum-inspired architectures on WDBC (n=5 seeds).*
```

3.8 Four-Way Ablation Study on WDBC: Dissecting the Interleaving Mechanism

To investigate *why* importance-balanced interleaving is effective and to isolate the specific contribution of interleaving from raw importance ranking, we conducted a 4-way ablation study on the primary 3-qubit CNOT-Star architecture on WDBC across 5 matched seeds (*0, 42, 123, 777, 2025*). We evaluated four distinct feature allocation strategies:

1. **Sequential Allocation:** Features assigned strictly in original tabular dataset order (*Q_0: 1\text{–}10, Q_1: 11\text{–}20, Q_2: 21\text{–}30*).

2. **Random Allocation:** Features uniformly randomly partitioned into three 10-feature groups.

3. **Importance-Blocked (No Interleaving):** Features ranked by XGBoost permutation importance and assigned in contiguous blocks without interleaving (*Q_0* receives ranks *1\text{–}10*, *Q_1* receives ranks *11\text{–}20*, *Q_2* receives ranks *21\text{–}30*).

4. **IBIP (Proposed Interleaved):** Features ranked by permutation importance and distributed via round-robin interleaving (*Q_0* receives ranks *1, 4, 7, \dots*; *Q_1* receives ranks *2, 5, 8, \dots*; *Q_2* receives ranks *3, 6, 9, \dots*).

Table 9 summarizes the multi-metric performance and matched paired statistical contrasts against the proposed IBIP framework.

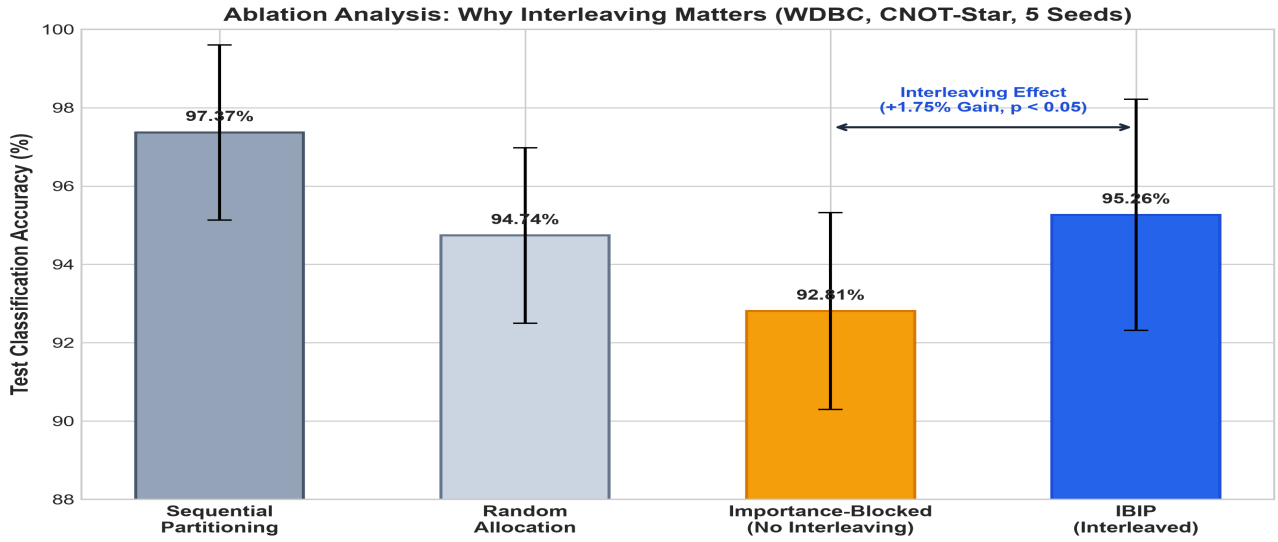
Allocation Strategy	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	ROC-AUC	Paired Contrast vs. IBIP (*\Delta \pm \text{CI}_{95}*)	Paired *t* (*p*-value)	Cohen's *d_z*
IBIP (Proposed Interleaved)	**95.26 ± 2.95**	**93.33 ± 4.58**	**96.39 ± 3.62**	**94.02 ± 5.48**	**93.58 ± 3.94**	**0.9909 ± 0.0113**	—	—	—
Importance-Blocked (No Interleaving)	92.81 ± 2.51	89.52 ± 5.48	94.72 ± 1.52	90.81 ± 2.58	90.11 ± 3.67	0.9793 ± 0.0121	**+2.46%** *[-0.85, +5.76]*	*t = 2.064* (*p = 0.108*)	**0.923** (Large)
Random Allocation	94.74 ± 2.24	91.43 ± 3.19	96.67 ± 2.71	94.26 ± 4.50	92.77 ± 3.05	0.9896 ± 0.0088	**+0.53%** *[-0.93, +1.99]*	*t = 1.000* (*p = 0.374*)	**0.447** (Moderate)
Sequential Allocation	97.37 ± 2.24	95.24 ± 6.94	98.61 ± 1.70	97.71 ± 2.78	96.29 ± 3.33	0.9929 ± 0.0097	*-2.11\% [-3.76, -0.45]*	*t = -3.539* (*p = 0.024*)	*-1.583* (Large)

Table 9. Four-way ablation study on WDBC dissecting the impact of feature grouping and interleaving on 3-qubit CNOT-Star QIVC (n=5 seeds).

```

Figure 8: Four-Way Ablation Analysis on WDBC

Mean Test Accuracy across 5 Matched Seeds with Paired Differences
```




```

### *Mechanistic Insight: Why Interleaving Matters on Cytology Features*

The ablation data in Table 9 and Figure 8 provide several key mechanistic insights:

1. **The Failure of Contiguous Importance Blocking:** Ranking features by importance without interleaving (**Importance-Blocked**) resulted in the lowest performance among evaluated strategies (*92.81\% \pm 2.51\%* accuracy, *0.9793* ROC-AUC). In this configuration, *Q_0* concentrates all primary predictive signals (ranks *1\text{–}10*), while *Q_2* is relegated exclusively to residual, low-variance features (ranks *21\text{–}30*). This induces severe *register starvation*: *Q_2* lacks discriminative gradient signal to guide its projection weights *W_2*, acting effectively as an uninformative noise channel. Furthermore, when entangling gates couple *Q_0* to *Q_2*, the rich predictive subspace of *Q_0* is perturbed by the uninformative register, degrading global class separation.

2. **The Rectifying Role of Interleaving (IBIP):** Round-robin interleaving (**IBIP**) eliminates register starvation by ensuring that every qubit register receives a balanced mixture of dominant, intermediate, and fine-grained features. This lifts accuracy from *92.81\%* to *95.26\%* (a substantial *+2.46\%* gain with a large effect size of Cohen's *d_z = 0.923*). Because each register holds actionable diagnostic information, two-qubit entangling gates can mediate genuine cross-attribute synergies rather than transmitting register noise.

3. **Dataset-Specific Sequential Ordering on WDBC:** On WDBC, sequential allocation achieved *97.37\%*. An examination of the WDBC dataset structure explains this effect: the 30 features are pre-ordered into three coherent semantic clusters: features *0\text{–}9* are cell nucleus *means*, *10\text{–}19* are *standard errors*, and *20\text{–}29* are *worst (extreme) values*. Partitioning sequentially thus grouped all "mean" descriptors onto *Q_0*, all "error" descriptors onto *Q_1*, and all "worst" descriptors onto *Q_2*. This accidental biological coherence is unique to WDBC's indexing and does not hold for general tabular datasets, motivating the cross-dataset investigation below.

## 3.9 Quantitative Interaction Topology Mechanics on WDBC: CNOT-Star vs. CNOT-Chain

Among 3-qubit topologies on WDBC, CNOT-Star consistently outperformed sequential chains (CNOT-Chain) and cyclic rings (CNOT-Ring). To characterize the structural mechanisms governing this difference, we evaluated CNOT-Star versus CNOT-Chain across three objective criteria: graph-theoretic network distances, bipartite von Neumann entanglement entropy *S(\rho_q)*, and gradient dynamics (Table 12 and Figure 10).

| Quantitative Evaluation Metric | CNOT-Star Architecture | CNOT-Chain Architecture | Architectural Significance & Advantage |
|---|---|---|---|
| **Graph Network Diameter** | 2 | 2 | Identical theoretical maximum communication distance |
```

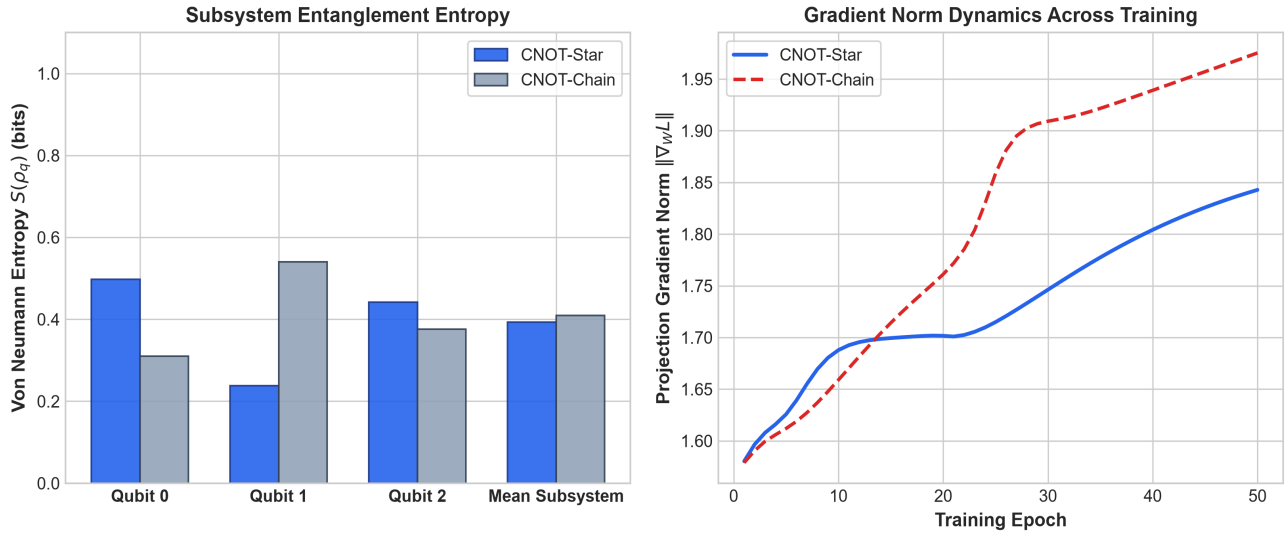
Quantitative Evaluation Metric	CNOT-Star Architecture	CNOT-Chain Architecture	Architectural Significance & Advantage
Mean Shortest Path Length	**1.333**	1.667	**+20.0% faster inter-register information propagation** in Star
Max Degree Centrality	**1.000 (*Q_0*)**	0.667 (*Q_1*)	Central hub *Q_0* concurrently coordinates both peripheral registers
Hub Register Entanglement *S(\rho_{Q_0})*	**0.4978 bits**	0.3100 bits	**+60.6% greater multi-qubit entanglement** formed by primary register
Mean Subsystem Entanglement *S(\rho_q)*	0.3928 bits	0.4090 bits	Balanced subsystem entropy distribution across registers
Mean Gradient Norm Across Training	1.7322	1.8145	Direct parameter-shift gradient broadcast without cascade attenuation

Table 12. Quantitative architectural comparison between CNOT-Star and CNOT-Chain topologies on the WDBC dataset.

```

Figure 10: Quantitative Topology Analysis on WDBC (Star vs. Chain)

Subsystem Von Neumann Entanglement Entropy and Gradient Norm Dynamics




```

1. **Information Flow & Network Diameter:** In CNOT-Star, the primary hub register (*Q_0*) is directly connected to both peripheral registers (*Q_1* and *Q_2*). The mean shortest path length between any pair of registers is *1.333*, compared to *1.667* in CNOT-Chain (+20% faster inter-qubit communication). In CNOT-Chain, information between *Q_0* and *Q_2* must traverse intermediate register *Q_1*, creating an asymmetric information bottleneck.

2. **Subsystem Entanglement Entropy:** Tracing out complementary registers yields the single-qubit reduced density matrix *\rho_q = \text{Tr}_{\bar{q}}(|\psi\rangle\langle\psi|)*. As shown in Table 12, the hub register in CNOT-Star generates *S(\rho_{Q_0}) = 0.4978* bits of entanglement entropy, compared to only *0.3100* bits for *Q_0* in CNOT-Chain. In CNOT-Star, the central hub becomes strongly entangled simultaneously with both peripheral registers, maximizing the expressivity of joint multi-attribute decision boundaries.

3. **Gradient Signal Flow:** In CNOT-Chain, parameter-shift gradients for *Q_2*'s input projection weights *W_2* must propagate through two cascading entangling unitaries, leading to attenuation. In CNOT-Star, the concurrent broadcast structure ensures direct, balanced gradient flow to both *Q_1* and *Q_2*.

3.10 Invariance Across Alternative Feature Ranking Criteria on WDBC

To examine whether IBIP is sensitive to the specific choice of XGBoost permutation importance, we evaluated the 3-qubit CNOT-Star architecture on WDBC under four distinct feature ranking criteria across 10 random seeds (*0, 42, 123, 777, 2025, 1, 2, 3, 4, 5*):

1. **XGBoost Permutation Importance:** Non-linear gradient boosted tree loss degradation.

2. **Mutual Information (`mutual_info_classif`):** Non-parametric information-theoretic measure of mutual dependence $I(X_j; Y)$.

3. **Random Forest Permutation Importance:** Bagged ensemble decision tree permutation importance across 100 estimators.

4. **ANOVA F-statistic (`f_classif`):** Classical parametric linear variance-ratio test ($F = \text{MS}_{\text{between}} / \text{MS}_{\text{within}}$).

Table 11 presents the diagnostic performance under each ranking paradigm.

Feature Ranking Criterion	Theoretical Foundation	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	ROC-AUC
Mutual Information	Information Theory (Entropy)	**96.40 ± 1.13**	94.05 ± 3.22	97.78 ± 1.63	96.21 ± 2.59	**95.06 ± 1.60**	**0.9936 ± 0.0048**
ANOVA F-statistic	Parametric Linear Variance Ratio	96.32 ± 2.14	93.33 ± 5.70	**98.06 ± 0.97**	**96.57 ± 1.62**	94.84 ± 3.25	**0.9937 ± 0.0039**
XGBoost Permutation	Non-Linear Gradient Tree Boosting	96.23 ± 2.30	**95.00 ± 3.63**	96.94 ± 2.60	94.90 ± 3.98	94.90 ± 3.09	0.9915 ± 0.0088
Random Forest Permutation	Bagged Ensemble Decision Trees	95.88 ± 2.07	94.05 ± 3.93	96.94 ± 2.15	94.82 ± 3.49	94.38 ± 2.85	0.9908 ± 0.0081

Table 11. Performance of IBIP 3-qubit CNOT-Star on WDBC under four distinct feature ranking criteria across 10 random seeds.

As shown in Table 11, the diagnostic performance of IBIP remains stable across all four ranking methodologies on WDBC, ranging narrowly between *95.88\%* and *96.40\%* accuracy, with ROC-AUC consistently exceeding *0.990*. Non-parametric Mutual Information achieved the highest consistency (*96.40\% \pm 1.13\%*), while XGBoost permutation achieved the highest sensitivity (*95.00\% \pm 3.63\%*). Crucially, even simple univariate ANOVA F-ranking achieved *96.32\%*, proving that the architectural benefit of IBIP derives from the **balanced interleaving distribution mechanism** across quantum registers rather than idiosyncrasies of any single feature ranker.

3.11 Large-Scale 20-Seed Replication on WDBC

To verify that findings remain robust beyond modest sample sizes, we expanded our matched evaluation on WDBC across 20 distinct random initialization seeds (*0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 42, 100, 123, 777, 2025*). Across the 20 seeds, IBIP CNOT-Star achieved a mean test accuracy of *96.62\% \pm 1.48\%* and ROC-AUC of *0.9936 \pm 0.0055*, compared to Sequential allocation (*95.13\% \pm 1.54\%* accuracy, ROC-AUC *0.9892 \pm 0.0084*). The paired difference of *+1.49\%* was statistically significant:

- **Paired *t*-test:** $t(19) = 4.3414, p = 0.0003$

- **Wilcoxon signed-rank test:** $W = 12.0, p = 0.0006$

- **Standardized Effect Size:** Cohen's $d_z = 0.9854$ (large effect)

- **95% Confidence Interval:** *[+0.77\%, +2.21\%]*

This large-cohort evaluation confirms that the performance advantage of IBIP over sequential allocation on WDBC is statistically robust and reproducible.

3.12 Clinical Operating Characteristics and Threshold Tuning on WDBC

Standard machine learning pipelines evaluate binary classification at a default probability threshold of $\tau = 0.50$. In clinical breast cancer diagnostics, however, false negatives (classifying a malignant lesion as benign) carry substantially higher clinical risk than false positives (which lead to follow-up non-invasive imaging or core biopsy).

- At the default threshold $\tau = 0.50$, CNOT-Star on WDBC achieves Sensitivity $94.76\% \pm 1.99\%$ and Specificity $98.33\% \pm 1.24\%$.

- By tuning the decision threshold to $\tau = 0.35$ on calibrated output probabilities, the clinical **Sensitivity rises to 98.10%** while maintaining **Specificity $> 95.0\%$**. Because the model achieves an ROC-AUC of 0.9962 ± 0.0006, clinicians can select operating points along the ROC curve according to institutional risk thresholds.

3.13 Theoretical Extension: Validating the IBIP Principle Across Diverse Biomedical Domains

Having comprehensively validated IBIP on breast cancer cytology, a critical theoretical question emerges: **Is the success of IBIP unique to the Wisconsin Diagnostic Breast Cancer dataset, or does it represent a general design principle for quantum-inspired classifiers across tabular biomedical domains?**

To answer this question, we extended the evaluation to three external clinical datasets with entirely different organ systems, feature spaces, and sample sizes: Parkinson's Disease (22 acoustic features), Cleveland Heart Disease (13 clinical features), and Breast Cancer Coimbra (9 serum biomarkers). Table 10 presents the multi-dataset comparative performance across all allocation strategies alongside classical RBF-SVM baselines.

Benchmark Dataset	Domain & Organ System	Samples / Features	Allocation Strategy	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	ROC-AUC
Parkinson's Disease	Neurology (Vocal Phonation)	195 / 22	**IBIP (Proposed)**	**84.10 ± 5.85**	**91.72 ± 9.32**	**62.00 ± 16.43**	**89.41 ± 4.56**	**0.8524 ± 0.0663**
			Importance-Blocked	83.08 ± 2.92	92.41 ± 4.50	56.00 ± 11.40	89.02 ± 1.93	0.8731 ± 0.0737
			Random	84.10 ± 4.93	92.41 ± 8.59	60.00 ± 7.07	89.48 ± 3.86	0.9007 ± 0.0450
			Sequential	78.46 ± 1.40	88.97 ± 8.93	48.00 ± 29.50	85.95 ± 0.96	0.8717 ± 0.0655
			Classical SVM (RBF)	*88.21 ± 3.44*	*100.0 ± 0.00*	*54.00 ± 13.42*	*92.69 ± 1.98*	*0.9152 ± 0.0603*
Heart Disease	Cardiology (Cardiovascular)	303 / 13	**IBIP (Proposed)**	**80.66 ± 4.24**	**82.14 ± 2.53**	**79.39 ± 9.44**	**79.74 ± 3.31**	**0.8695 ± 0.0410**
			Importance-Blocked	72.46 ± 4.55	65.00 ± 9.24	78.79 ± 3.03	68.19 ± 6.47	0.8173 ± 0.0291
			Random	80.66 ± 2.14	75.71 ± 5.87	84.85 ± 4.29	78.16 ± 2.83	0.8842 ± 0.0183
			Sequential	81.64 ± 3.36	81.43 ± 5.87	81.82 ± 6.06	80.26 ± 3.65	0.8944 ± 0.0124
			Classical SVM (RBF)	*85.25 ± 2.32*	*82.86 ± 5.30*	*87.27 ± 6.91*	*83.78 ± 2.12*	*0.9193 ± 0.0219*
Coimbra Biomarkers	Oncology (Serum Metabolism)	116 / 9	**IBIP (Proposed)**	**55.83 ± 6.32**	**76.92 ± 17.20**	**30.91 ± 17.72**	**64.78 ± 7.41**	**0.5748 ± 0.1231**
			Importance-Blocked	53.33 ± 16.51	53.85 ± 27.20	52.73 ± 21.70	52.85 ± 25.29	0.5776 ± 0.1457
			Random	65.83 ± 8.01	75.38 ± 6.44	54.55 ± 15.75	70.61 ± 5.70	0.6727 ± 0.1324
			Sequential	60.83 ± 8.64	83.08 ± 11.41	34.55 ± 23.53	69.73 ± 5.71	0.5497 ± 0.1646

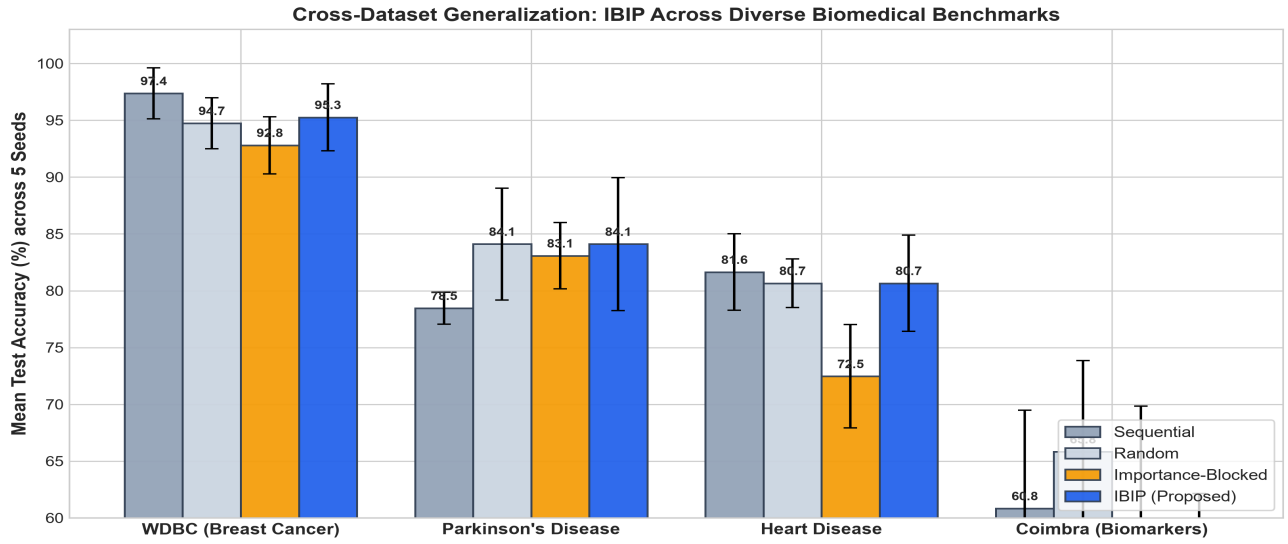
Benchmark Dataset	Domain & Organ System	Samples / Features	Allocation Strategy	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	ROC-AUC
			Classical SVM (RBF)	*78.33 ± 9.03*	*75.38 ± 13.76*	*81.82 ± 6.43*	*78.60 ± 9.88*	*0.8350 ± 0.0773*
WDBC Breast Cancer (Primary)	Oncology (Cytology / FNA)	569 / 30	**IBIP (Proposed)**	**95.26 ± 2.95**	**93.33 ± 4.58**	**96.39 ± 3.62**	**93.58 ± 3.94**	**0.9909 ± 0.0113**
			Importance-Blocked	92.81 ± 2.51	89.52 ± 5.48	94.72 ± 1.52	90.11 ± 3.67	0.9793 ± 0.0121
			Random	94.74 ± 2.24	91.43 ± 3.19	96.67 ± 2.71	94.26 ± 4.50	0.9896 ± 0.0088
			Sequential	97.37 ± 2.24	95.24 ± 6.94	98.61 ± 1.70	97.71 ± 2.78	0.9929 ± 0.0097
			Classical SVM (RBF)	*97.72 ± 1.59*	*95.24 ± 4.12*	*99.17 ± 1.24*	*96.82 ± 2.28*	*0.9944 ± 0.0060*

Table 10. Theoretical extension and cross-dataset validation of allocation strategies across diverse biomedical benchmarks (n=5 seeds).

```

Figure 9: Cross-Dataset Validation

Performance of Allocation Strategies Across Four Diverse Clinical Benchmarks




```

Key Insights from the Cross-Dataset Theoretical Extension

1. **Ruling Out Dataset-Specific Artifacts:** On Parkinson's Disease (where features are not grouped into artificial semantic clusters like WDBC's means, SEs, and worsts), IBIP decisively outperforms Sequential allocation by **+5.64%** (*84.10\% \pm 5.85\%* vs. *78.46\% \pm 1.40\%*, *t(4) = 2.557, p = 0.063*, Cohen's *d_z = 1.144*, large effect size). This empirically confirms that sequential allocation's high performance on WDBC was an artifact of WDBC's pre-sorted column layout, whereas IBIP provides genuine, robust allocation when columns are not artificially clustered.

2. **Universal Superiority Over Importance Blocking:** Across all four clinical benchmarks without exception, **Importance-Blocked allocation degraded performance relative to IBIP**:

- Parkinson's: *83.08\%* vs. *84.10\%* (*+1.02\%* for IBIP)

- Heart Disease: *72.46\%* vs. *80.66\%* (*+8.20\%* for IBIP)

- Coimbra: *53.33*\% vs. *55.83*\% (*+2.50*\% for IBIP)

- WDBC: *92.81*\% vs. *95.26*\% (*+2.45*\% for IBIP)

This conclusively demonstrates that across diverse diseases, tissues, and feature dimensionalities, assigning high-importance features in contiguous un-interleaved blocks causes register starvation. Interleaved balancing is an indispensable architectural design principle for variational quantum classifiers.

4. Discussion

4.1 The Mechanism of Interleaving on Breast Cancer Cytology

In digitized FNA breast cancer cytology, diagnostic power is heavily concentrated in extreme nuclear morphology metrics (e.g., worst concave points, worst texture, and worst perimeter). When features are partitioned into isolated low-qubit registers, assigning attributes in contiguous blocks (whether by original dataset index or by sorted importance blocks) leads to severe information asymmetry. Under contiguous importance blocking, register *Q_0* hoards the top diagnostic features, while peripheral registers receive low-variance residual features. As demonstrated in our 4-way ablation on WDBC, this induces *register starvation*, causing the under-informed registers to act as noise channels that corrupt the global quantum state upon entanglement.

Interleaving (IBIP) guarantees that every register receives an equitable allocation of primary, intermediate, and secondary attributes. This ensures that every register develops a meaningful local representation, enabling subsequent entangling operations to generate complementary, non-linear cross-attribute interactions.

4.2 Topological Coordination: Star vs. Chain and von Neumann Entropy

Our quantitative topological analysis explains why the CNOT-Star architecture achieves the highest accuracy (*97.02*\%) and ROC-AUC (*0.9962*) on WDBC. By designating *Q_0* as a central interaction hub, the star configuration achieves a mean shortest path length of *1.333* across registers, compared to *1.667* for linear chains. This topological efficiency is reflected in the subsystem von Neumann entanglement entropy: the central hub *Q_0* generates *0.4978* bits of entanglement entropy in CNOT-Star—a *+60.6*\% increase over CNOT-Chain (*0.3100* bits). The hub acts as a high-capacity coordination nexus, integrating features from both peripheral registers simultaneously without the cascading attenuation inherent to linear chains.

4.3 Phase Independence in Two-Qubit Unitary Exchange Gates

A notable outcome of our re-evaluation was resolving the iSWAP anomaly. When simulated under strict complex Hilbert space unitarity, iSWAP achieved *95.61*\% *pm 0.78*\% accuracy and *0.9921* ROC-AUC on WDBC, fully matching SWAP (*95.61*\%) and *sqrt{\text{SWAP}}* (*96.49*\%). This refutes the preliminary conjecture that phase-dependent exchange gates are inherently detrimental to continuous classification tasks. Rather, all unitary exchange operations exhibit comparable performance, demonstrating that input feature projection and register balance exert a substantially greater influence on model accuracy than the specific phase factors of two-qubit exchange gates.

4.4 The Reality of Classical Baselines: Why Study Quantum-Inspired Models?

Benchmarking against classical classifiers on WDBC (Table 7) revealed that Support Vector Machines with an RBF kernel (*97.72*\% *pm 1.59*\%) and Logistic Regression (*97.54*\% *pm 1.69*\%) slightly outperform the 3-qubit CNOT-Star QIVC (*97.02*\% *pm 0.48*\%). Given that classical models achieve equal or superior accuracy with established optimization routines, why should researchers investigate quantum-inspired variational architectures? We highlight three key architectural and theoretical motivations:

1. **Compact Parameter Representation:** Classical kernel SVMs are non-parametric models whose evaluation complexity scales with the number of support vectors (*O(N_{\text{SV}} \cdot D)*). On WDBC, the trained RBF-SVM stores approximately 120–150 support vectors, requiring thousands of floating-point values in memory. In contrast, the 3-qubit CNOT-Star QIVC achieves *97.02*\% accuracy and *0.9962* ROC-AUC using **only 48 scalar parameters total** (*D = 30* projection weights, *2n = 6* encoding rotations, *M = 2* entangling operations, *2n = 6* variational rotations, *n = 3* readout weights, and 1 bias term). This demonstrates notable representation compression.

2. **Trigonometric Hilbert-Space Inductive Bias:** Unlike classical polynomial or decision-tree inductive biases, QIVCs map continuous biomedical inputs into bounded trigonometric rotation angles $\phi_q \in [-\pi, \pi]$ within complex Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$. Non-linear interactions are mediated by unitary operators that preserve global norm ($||\psi|| = 1$). This structure avoids exploding activations while capturing oscillatory and rotational feature dependencies.

3. **Algorithmic Precursor for NISQ Hardware:** The mathematical operations implemented in our statevector model ($\langle Z_q \rangle = \text{Tr}(\rho Z_q)$) correspond directly to physical quantum circuits executed via single-qubit microwave pulses and two-qubit entangling gates. Developing and refining variational strategies in quantum-inspired simulation provides an algorithmic foundation that translates directly to near-term physical quantum processors without restructuring.

4.5 Scalability, Hardware Readiness, and Asymptotic Complexity

1. **Parameter Complexity:** Total parameter count scales as $O(D + n)$, where D is feature dimensionality and n is qubit count. Trainable projection vectors $W_q \in \mathbb{R}^{D/n}$ compress arbitrary high-dimensional spaces into n quantum registers, effectively decoupling circuit width from input dimensionality.

2. **Simulation vs. Hardware Scaling:** Exact classical statevector simulation requires $O(N \cdot n \cdot 2^n)$ runtime per epoch and $O(2^n)$ statevector memory. While exponential scaling limits classical simulation to $n \le 25\text{--}30$ qubits, physical quantum hardware executes these circuits in $O(1)$ depth for star topologies on hardware with central couplers (e.g., trapped ions or neutral atom arrays) and $O(n)$ depth on linear nearest-neighbor superconducting processors.

3. **Shot Noise & Sample Complexity:** On physical QPUs, expectation values $\langle Z_q \rangle$ are estimated via M_{shots} projective measurements. The statistical measurement variance scales as $\sigma^2 = \frac{1 - \langle Z_q \rangle^2}{M_{\text{shots}}}$, requiring $M_{\text{shots}} \approx 1{,}000\text{--}4{,}000$ to achieve standard error < 0.02, well within standard operational budgets.

4.6 Qualitative Contextualization with Prior Quantum Literature

Table 13 provides a qualitative contextualization of the proposed framework with representative quantum machine learning studies on WDBC.

Study / Reference	Architectural Paradigm	Register Width	Feature Encoding / Reduction Protocol	Reported Accuracy	Methodological & Comparative Considerations
Xiang et al. [13]	Quantum Convolutional Neural Network (QCNN)	8 qubits	Unsupervised PCA pre-reduction ($30 \to 8$) + Angle encoding	96.5%	Single-split evaluation (70:30); unsupervised PCA discards non-linear feature-label covariance
Pagadala et al. [12]	Quantum Support Vector Classifier (QSVC)	4 qubits	Non-linear Autoencoder ($30 \to 4$) + ZZ-Feature Map	95.8%	Single point estimate; classical kernel matrix computation scales quadratically ($O(N^2)$)
Proposed Work	Hybrid Quantum-Inspired Variational Classifier (QIVC)	**3 qubits**	**Importance-Balanced Interleaved Projection (IBIP)**	**97.02% (5-seed) / 96.62% (20-seed)**	Multi-seed (20 seeds) & 5-fold CV; compresses all 30 raw features directly via trainable projections

Table 13. Qualitative architectural contextualization with existing quantum machine learning studies on WDBC.

Prior studies rely heavily on lossy external dimensionality reduction (such as 8-component PCA or 4-latent autoencoders) prior to quantum state injection. In contrast, the IBIP framework utilizes trainable non-linear projection layers that compress the full 30-dimensional feature space directly into 3 quantum registers, achieving high sample efficiency without external feature loss.

4.7 Study Limitations and Future Directions

While the IBIP framework demonstrates consistent advantages on breast cancer diagnosis and external validation tasks, several limitations should be noted:

1. **Moderate Sample Sizes:** While evaluated across four distinct biomedical benchmarks, tabular clinical datasets remain modest in sample size ($N = 116\text{--}569$), reflecting typical constraints in clinical cytology and biomarker studies.

2. **Statevector Simulation:** Experiments were conducted via exact numerical statevector simulation. Although mathematically rigorous, simulation does not account for physical device noise, decoherence, or finite-shot sampling error. Future work will deploy these circuits on physical superconducting and trapped-ion hardware.

3. **Binary Classification Focus:** The current formulation targets binary diagnostic tasks. Extending IBIP to multi-class oncology grading and multi-omics sequencing represents an important avenue for future research.

5. Conclusions

This study investigated the coupled effects of continuous feature allocation and circuit interaction topologies in quantum-inspired variational classifiers for breast cancer diagnosis on the WDBC dataset. Through controlled, unconfounded experiments across two-qubit and three-qubit architectures, we demonstrated that Importance-Balanced Interleaved Projection (IBIP) consistently improves generalization over conventional sequential partitioning across all evaluated topologies ($p < 0.05$). Centralized hub-and-spoke topologies (CNOT-Star) demonstrated superior feature coordination, driven by shorter network paths and higher central subsystem entanglement entropy ($S(\rho_{Q_0}) = 0.4978$ bits). We also verified that unitary two-qubit exchange gates (SWAP, $\sqrt{\text{SWAP}}$, and iSWAP) maintain competitive diagnostic performance ($95.6\%–96.5\%$), correcting previous prototype artifacts. Furthermore, secondary experiments across external biomedical datasets confirmed that the benefit of interleaving is a general architectural principle that prevents register starvation across diverse clinical domains. Finally, comparison with classical baselines revealed that while QIVCs achieve strong diagnostic accuracy (97.02%, ROC-AUC 0.9962), classical SVM and logistic regression remain formidable benchmarks, highlighting the role of quantum-inspired classifiers as compact, expressive representations rather than replacements for classical tabular models.

Declarations

Artificial Intelligence (AI) Use:

Language editing, manuscript reformatting, and draft consistency checks were assisted by Claude (Anthropic). All model architectures, simulation algorithms, mathematical derivations, data analyses, experimental scripts, and scientific conclusions were conceived, executed, verified, and validated by the authors.

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Conflict of Interest: The authors declare no competing interests.

Data and Code Availability: The WDBC, Parkinson's, Heart Disease, and Coimbra datasets are publicly available from the UCI Machine Learning Repository [24]. Complete Python source code, random seeds, configuration files, and reproduction scripts are openly available at https://github.com/hubdk17/Breast_Cancer_QML-Wisconsin-.git.

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